

# Benjamin T. James

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## Overview

Computational biologist and engineer (MIT EECS Ph.D., 2026) - two co-first-author Cell papers, contributor to Terminal-Bench Science. Building scalable numerical infrastructure and agent-evaluation tooling that lets computational biology meet AI at the scale real datasets demand in service of concrete biological discoveries.

## Education

### • Massachusetts Institute of Technology

September 2019 - August 2026

Ph.D in Electrical Engineering and Computer Science, minor in Econometrics

Cambridge, Massachusetts

◦ GPA: 4.80/5.00

### • University of Tulsa

Aug 2015 - May 2019

BSc in Mathematics, BSc in Computer Science

Tulsa, Oklahoma

◦ GPA: 3.747/4.000

## Selected Research

 Includes two (co-) first author Cell papers, one second-author Nature paper

### • Single-cell dissection of the human brain

2019 - 2026

Tools: R, Python, Bash, C++

[\[🔗\]](#)

◦ Integrated high-dimensional, multimodal (scRNA-seq/scATAC-seq) data from **hundreds of postmortem human brains** and mouse brains from corresponding model systems

◦ Analyzed these data across several **human disease contexts** and brain regions using **functional genomics**, careful integration/atlas, and case-control causal inference to infer disease pathology.

◦ Led computational analysis on a cross-species striatum atlas (670K nuclei, 69 human donors) defining rare subpopulations including molecular subtypes of the century-old islands of Calleja, a notable species-conserved dorsolateral-ventromedial gradient, and human-enriched druggable targets invisible to rodent models, notably CHRM5 (a Cofenyl/xanomeline receptor) and GLP-1R/GIPR sites with CNS mechanistic implications for obesity and addiction therapeutics.

◦ Created new methods to make inferences on new technologies (e.g. spatial transcriptomics) and gene module discovery, mostly using random matrix theory and high-dimensional statistics

◦ Connected millions of DNA elements active across human tissues to their target genes

◦ Linville\*, R. M., **James\***, B. T., Galani, K., Ho, L., Shin, J., Oliver, E., Bock, R., Murray, E. M., Louca, M., Oke, O., Wang, C., Engelberg-Cook, E., DeTure, M., Farrell, V., Fitzwalter, B., Pineda, S., Sathitloetsakun, S., Liang, X., Madras, B., Dickson, D. W., Mash, D. C., Turecki, G., Wheeler, V. C., Alvarez, V. A., Gabuzda, D., Kellis, M., Heiman, M. (2026). Cross-species single cell atlas of the striatum defines cell type and subregion disease vulnerabilities. *Cell*, in press.

◦ Xiong\*, X., **James\***, B. T., Boix\*, C. A., Park, Y. P., Galani, K., Victor, M. B., Sun, N., Hou, L., Ho, L.-L., Mantero, J., Scannail, A. N., Dileep, V., Dong, W., Mathys, H., Bennett, D. A., Tsai, L.-H., & Kellis, M. (2023). Epigenomic dissection of Alzheimer's disease pinpoints causal variants and reveals epigenome erosion. *Cell*, 186(20), 4422-4437.e21. <https://doi.org/10.1016/j.cell.2023.08.040>

◦ Boix, C. A., **James, B. T.**, Park, Y. P., Meuleman, W., & Kellis, M. (2021). Regulatory genomic circuitry of human disease loci by integrative epigenomics. *Nature*, 590(7845), 300-307. <https://doi.org/10.1038/s41586-020-03145-z>

### • Intelligent clustering of DNA sequences

2016-2019

Tools: C++/OpenMP; gdb/valgrind

[\[🔗\]](#)

◦ Developed a fast, concurrent method as an undergraduate student to approximate an expensive method for DNA similarity (alignment) and used for clustering. Led to 5 journal publications as an undergrad.

◦ **James, B. T.**, Luczak, B. B., & Girgis, H. Z. (2018). MeShClust: an intelligent tool for clustering DNA sequences. *Nucleic Acids Research*, 46(14), e83. <https://doi.org/10.1093/nar/gky315>

## Skills

• **Programming Languages:** R and Python (Daily use), C++ (+pybind11/Rcpp), Golang, Bash/POSIX sh, C, AWK & Perl, Julia, Common Lisp

• **Research-specific Packages and Tools:** Bioconductor, scverse, samtools/bedtools, Eigen, NumPy/SciPy/SK-Learn, Snakemake & Nextflow, brms

• **DevOps/MLOps:** De-facto manager of two lab-wide compute clusters, managing software (Lmod, conda, Docker, singularity environments, CUDA/NV admin, Terraform, Ansible, etc). Experienced Linux user (Debian/Gentoo user since 2014), see [my dotfiles](#), formerly an ArchLinux [maintainer](#)

• **Side projects:** Contributor (under review) to **terminal-bench-science** [🔗](#), [🔗](#) - benchmark for AI agents on scientific workflows; parent benchmark is featured on frontier model cards, Wrote an **agentic** harness tool to specify disposable agent virtual machines [🔗](#) and deploy them [🔗](#), Contributed the dot-product distance into the [RcppAnnoy](#) package required by R genomics libraries [🔗](#), Wrote the sparse-matrix math for the [psychromVAR](#) package for transcription factor enrichment in snATAC-seq [🔗](#), wrote a complete BF compiler [🔗](#) and a Pascal compiler frontend [🔗](#) both in C, wrote and deployed a multi-user webapp on a VPS [🔗](#) (Apache/Flask/PostgreSQL)

## Honors and Awards

### • NSF GRFP

Awarded May 2019

National Science Foundation

### • Cronin Fellowship

Awarded April 2019

MIT EECS department